In [2]:
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class Node:
    """Node representing a single student record."""
    def __init__(self, roll_no, name, marks):
        self.roll_no = roll_no
        self.name = name
        self.marks = marks
        self.next = None


class StudentLinkedList:
    """Singly Linked List to manage student records."""
    def __init__(self):
        self.head = None

    def add_student(self, roll_no, name, marks):
        """Add a new student record at the end."""
        new_node = Node(roll_no, name, marks)
        if self.head is None:
            self.head = new_node
        else:
            temp = self.head
            while temp.next:
                temp = temp.next
            temp.next = new_node
        print(f"Student {name} added successfully!")

    def delete_student(self, roll_no):
        """Delete a student record by Roll Number."""
        temp = self.head
        prev = None

        while temp:
            if temp.roll_no == roll_no:
                if prev:
                    prev.next = temp.next
                else:
                    self.head = temp.next
                print(f"Record with Roll No {roll_no} deleted successfu
                return
            prev = temp
            temp = temp.next

        print(f"Record with Roll No {roll_no} not found!")

    def update_student(self, roll_no, new_name=None, new_marks=None):
        """Update a student's name or marks."""
        temp = self.head
        while temp:
            if temp.roll_no == roll_no:
                if new_name:
                    temp.name = new_name
                if new_marks is not None:
                    temp.marks = new_marks
                print(f"Record with Roll No {roll_no} updated successfu
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                return
            temp = temp.next
        print(f"Record with Roll No {roll_no} not found!")

    def search_student(self, roll_no):
        """Search for a student by Roll Number."""
        temp = self.head
        while temp:
            if temp.roll_no == roll_no:
                print(f"Found -> Roll No: {temp.roll_no}, Name: {temp.n
                return
            temp = temp.next
        print(f"Record with Roll No {roll_no} not found!")

    def sort_records(self, key="roll_no", ascending=True):
        """Sort records based on Roll Number or Marks."""
        if self.head is None or self.head.next is None:
            return

        # Convert linked list to list for sorting
        nodes = []
        temp = self.head
        while temp:
            nodes.append(temp)
            temp = temp.next

        reverse = not ascending
        if key == "marks":
            nodes.sort(key=lambda x: x.marks, reverse=reverse)
        else:
            nodes.sort(key=lambda x: x.roll_no, reverse=reverse)

        # Rebuild linked list
        self.head = nodes[0]
        temp = self.head
        for node in nodes[1:]:
            temp.next = node
            temp = node
        temp.next = None

        print(f"Records sorted by {key} in {'ascending' if ascending el

    def display(self):
        """Display all student records."""
        if self.head is None:
            print("No records to display!")
            return

        print("\nStudent Records:")
        print("Roll No\tName\tMarks")
        print("-" * 25)
        temp = self.head
        while temp:
            print(f"{temp.roll_no}\t{temp.name}\t{temp.marks}")
            temp = temp.next
        print("-" * 25)
```

```
# ---------------- Example Usage ----------------
if __name__ == "__main__":
    s_list = StudentLinkedList()
    s_list.add_student(101, "Amit", 85)
    s_list.add_student(102, "Priya", 92)
    s_list.add_student(103, "Rohan", 78)
    s_list.display()
    s_list.update_student(102, new_marks=95)
    s_list.search_student(103)
    s_list.sort_records(key="marks", ascending=False)
    s_list.display()

    s_list.delete_student(101)
    s_list.display()
```

```
Student Amit added successfully!
Student Priya added successfully!
Student Rohan added successfully!

Student Records:
Roll No Name    Marks
-------------------------
101     Amit    85
102     Priya   92
103     Rohan   78
-------------------------
Record with Roll No 102 updated successfully!
Found -> Roll No: 103, Name: Rohan, Marks: 78
Records sorted by marks in descending order!

Student Records:
Roll No Name    Marks
-------------------------
102     Priya   95
101     Amit    85
103     Rohan   78
-------------------------
Record with Roll No 101 deleted successfully!

Student Records:
Roll No Name    Marks
-------------------------
102     Priya   95
103     Rohan   78
-------------------------
```